VAISHNAVI KOLAPE

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Security operations focused cyber security graduate (GCHQ and NCSC certified MSc) with 2+ years resolving severe, business critical incidents on live payment systems within **SLA** at Tata Trent, part of the Tata Group. Built and tested detection hands on: a **Splunk SIEM lab** across 5+ hosts and 8 attack scenarios, triaging alerts from Windows Event IDs 4625, 5156 and 5157, validating them against **Wireshark packet captures** and ruling out false positives, plus a **Google Nest Cam penetration test** that validated a **token replay vulnerability**. Proficient in **Splunk, SQL, MITRE ATT&CK, Microsoft Sentinel, Defender for Endpoint and Security Copilot**, with strong competencies in **alert triage, investigation, runbooks** and clear escalation across technical teams.

KEY SKILLS

- **Detection and Response:** alert triage, security event investigation, incident response, escalation of high risk findings, alert validation, identifying false positives and feeding back to improve detection logic and coverage, alert enrichment using packet captures and Windows event context, root cause analysis, guided response and documentation
- **Log Analysis and Telemetry:** Splunk searches, dashboards and alerts (lab project), SQL and SQL Plus queries, Windows Event ID analysis, correlating endpoint, identity, network and application logs, Wireshark, application log analysis at Trent
- **SIEM, EDR and Cloud:** Splunk (lab project), Microsoft Sentinel, Microsoft Defender for Endpoint, Microsoft Security Copilot and CrowdStrike Falcon (courses and certifications), Active Directory identity systems, Microsoft Azure, cloud security fundamentals (projects and certifications)
- **Threat Knowledge:** MITRE ATT&CK, attacker tactics and techniques, threat intelligence (CFC Underwriting workshop, Career Essentials in Cybersecurity), reconnaissance and brute force simulation with Kali Linux, TryHackMe labs
- **Runbooks and Automation:** investigation guides, runbooks, Python and PowerShell scripting, AI assisted tools including Microsoft Security Copilot (courses and certifications) for analysis and scripting, learning SOAR, benchmarking AI models (MSc dissertation)
- **Frameworks and Systems:** NIST CSF, ISO 27001, PCI DSS, GDPR, Windows Server, Linux, TCP/IP, firewalls, security patching
- **Communication and Teamwork:** clear updates and collaboration during investigations, cross team communication, collaborating with cross functional teams, documenting findings clearly, willingness to learn and take direction
- **Languages:** English, Hindi and Marathi (fluent); understands most Indian languages and Sanskrit
- **Core Skills:** Microsoft Office (Word, Excel, PowerPoint), analytical mindset, attention to detail, organisational skills, structured problem solving, proactive

PROFESSIONAL EXPERIENCE

Tata Trent (Tata Group), Junior Executive, IT Application Support and Customer Operations | Jun 2023 to Sep 2025 | Mumbai, India

Partners: SAP, network and vendor teams

- Investigated severe, business critical incidents on live POS and payment systems, using **application log analysis** and **root cause analysis** to meet **SLA** timelines
- Escalated complex issues to SAP, network and vendor teams with clear evidence and documented each outcome
- Monitored **endpoints** and system stability, checked **network** connectivity with ping and telnet, and supported **firewall** configurations
- Applied **security patching** and software upgrades on **Windows Server and Linux**, and supported the migration of on premises systems to **Microsoft Azure**
- Wrote **runbooks** and tracked incidents in **ManageEngine ServiceDesk**, keeping responses consistent
- Queried and reported data with **SQL Plus**, **Power BI**, **SAP data cubes** and **Excel**

PROJECTS

SIEM Based Threat Detection Lab | University of Birmingham | 2026

- Built a **Splunk** lab monitoring **5+ hosts** across **8 attack scenarios**, simulating reconnaissance and brute force attacks with Kali Linux and triaging the resulting alerts, validating each one and ruling out false positives against packet

captures

- Analysed Windows Event IDs **4625, 5156 and 5157** and validated attack traffic with Wireshark **packet captures** to establish what happened
- **Tech Stack:** Splunk, Kali Linux, Wireshark, Windows Event Logs
- **Key Results: dashboards** and alerts for suspicious login attempts and firewall activity across 8 attack scenarios

Privacy Preserving Document Processing (MSc Dissertation) | University of Birmingham | 2026

- Engineered a modular **Python** pipeline that detects and redacts sensitive information in corporate documents, with a security gateway that enforces block or allow decisions
- **Tech Stack:** Python, Microsoft Presidio, spaCy, Hugging Face transformers, Streamlit, GitLab
- **Key Results: 0.978 recall** for rule based detection; benchmarked four AI models and found the generative rival cost about **15x latency and 3x memory** for a gain within confidence intervals
- **Code:** github.com/vaishnavi-kolape/Project-1

Penetration Test, Google Nest Cam (team of five) | University of Birmingham | 2026

- Uncovered and validated a **token replay vulnerability** through traffic interception and mobile app analysis
- **Tech Stack:** Wireshark, Burp Suite, Nmap, Frida, JADX

BEng Final Year Project, OCR Text Detection using NLP | Ramrao Adik Institute of Technology | 2023

- Developed an OCR text detection system for local boards using **NLP**
- **Tech Stack:** Python, NLP, OCR

KEY ACHIEVEMENTS

- **Splunk SIEM lab** covering 8 attack scenarios across 5+ hosts, with dashboards and alerts for suspicious logins (2026)
- **0.978 recall** from the rule based detection layer of the MSc dissertation pipeline (2026)
- **Token replay vulnerability** identified and validated in a Google Nest Cam penetration test, in a team of five (2026)

EDUCATION

- **MSc Cyber Security** | University of Birmingham, UK | Sep 2025 to Sep 2026 GCHQ and NCSC certified. Modules: Incident Response; Malware and Pentesting Forensics; Network Security and Cryptography; Designing and Managing Secure Systems; Secure Software and Hardware Systems; Dependable Distributed Systems.
- **BEng Computer Engineering** | Ramrao Adik Institute of Technology, University of Mumbai, India | Jul 2019 to Jun 2023

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- **Career Essentials in Cybersecurity** (Microsoft and LinkedIn Learning) and **Cybersecurity Foundations** (LinkedIn Learning)
- **CFC Underwriting Technical Cyber Security Workshop** | threat analysis, threat intelligence, network security and incident response
- **Peer Coaching and Communication Skills** (Elliott Training and Consultancy, Accredited, 2026)
- **Leading in Engineering and Science Workshop** (University of Birmingham, Jun 2026)

LEADERSHIP AND COMMUNICATION

- **Event Head, CSI RAIT** (2023) | directed a team of 10 to deliver 3 technical placement seminars for 500+ students
- **Campus Leader, Techlearn.Live** (2023) | recruited and onboarded 50+ students, lifting campus engagement by 30% in one semester
- **Student Ambassador, University of Birmingham** (2026) | guided prospective students and families and answered enquiries at open days
- **Writer, Medium cyber security blog** | beginner friendly articles on WiFi security, TLS, VMs and sandboxing, ransomware and Zero Trust